

Perform the analysis of variance on the following and analyze the treatments using the LSD, Tukey's HSD, Duncan MRT, Scheffe's test with $\alpha = 0.05$.

Blocks	Treatments					
	1	2	3	4	5	6
1	1	3	6	4	3	2
2	1	4	4	8	5	1
3	3	6	7	8	4	3
4	2	3	2	3	2	1

1) Formulation of hypothesis:

$$H_0: \mu_1 = \mu_2 = \mu_3 = \mu_4 = \mu_5 = \mu_6$$

H_1 : At least one treatment means are not equal.

2) Level of significance: $\alpha = 0.05$

3) Test Statistics:

4) Calculation:

Blocks	Treatments						
	1	2	3	4	5	6	
1	1	3	6	4	3	2	19
2	1	4	4	8	5	1	23
3	3	6	7	8	4	3	31
4	2	3	2	3	2	1	13
Total	7	16	19	23	14	7	86
Mean	1.75	4	4.75	5.75	3.5	1.75	

$$C.F = T^2/n = (86)^2/24$$

$$C.F = 308.17$$

$$\text{Total SS} = \sum_i \sum_j X_{ij}^2 - C.F = 412 - 308.17$$

$$\text{Total SS} = 103.83$$

$$\text{B/w treatments: } \sum_j T_j^2/c - C.F = 360 - 308.17$$

$$\text{B/w treat} = 51.83$$

$$\text{B/w block SS} = \sum_i T_i^2/c - F = 336.67 - 308.17$$

$$\text{B/w block} = 28.50$$

$$\text{Error SS} = \text{Total SS} - \text{B/w treat} - \text{B/w block}$$

$$\text{Error SS} = 23.50$$

Anova table:

S.O.V	d.f	S.S	M.S	F-ratio
B/w Treat.	5	51.83	10.37	$F_1 = \frac{10.37}{1.57} = 6.61$
B/w Block	3	28.50	9.50	
Error	15	23.50	1.57	
Total	23	103.83		

5) Critical region: $F \geq F_{\alpha}(V_1, V_2)$

6) Conclusion: $F \geq F_{0.05}(5, 15)$, $F \geq 2.90$

We reject H_0 , we therefore apply the possible multiple comparison test to find out which mean differs from each other.

i) **LSD test:**

$$LSD = t_{\alpha/2}(v) \sqrt{\frac{2(MSE)}{r}} = \sqrt{ME \left(\frac{1}{r_i} + \frac{1}{r_j} \right)} \quad \text{for unequal case.}$$

Where "MSE" is the error (within) Mean square, "r" is the size of equal samples.

" $t_{\alpha/2}(v)$ " is the value of t at $\alpha/2$ level taken against the error degrees of freedom (v).

→ It is interesting to note that the LSD test is actually a two sample t-test.

→ It is customary to arrange the sample means in ascending order of magnitude. i.e. ~~$\bar{x}_1, \bar{x}_2, \bar{x}_3, \bar{x}_4, \bar{x}_5, \bar{x}_6$~~

Now we arrange the samples mean in ascending order.

$\bar{x}_1$

1.75

→ The least significant Difference is given by the selection

$$LSD = t_{0.025}(15) \sqrt{\frac{2 \times 1.57}{4}} = 1.89$$

→ Arrange the treatment means in ascending order.

$\bar{x}_6$	$\bar{x}_5$	$\bar{x}_4$	$\bar{x}_3$	$\bar{x}_2$	$\bar{x}_1$	$\bar{x}_6$	$\bar{x}_5$	$\bar{x}_4$	$\bar{x}_3$	$\bar{x}_2$	$\bar{x}_1$
5.75	4.75	4	3.50	1.75	1.75	1.75	3.50	4	4.75	5.75	

1 vs 6	$1.75 - 1.75 = 0 < 1.89$	Not significant
1 vs 5	$3.50 - 1.75 = 1.75 < 1.89$	Not sig.
1 vs 2	$4 - 1.75 = 2.25 > 1.89$	Significant
1 vs 3	$4.75 - 1.75 = 3 > 1.89$	Significant
1 vs 4	$5.75 - 1.75 = 4 > 1.89$	Significant
6 vs 5	$3.50 - 1.75 = 1.75 < 1.89$	Not sig.
6 vs 2	$4 - 1.75 = 2.25 > 1.89$	Significant
6 vs 3	$4.75 - 1.75 = 3 > 1.89$	Significant
6 vs 4	$5.75 - 1.75 = 4 > 1.89$	Significant
5 vs 2	$4 - 3.50 = 0.50 < 1.89$	Not sig.
5 vs 3	$4.75 - 3.50 = 1.25 < 1.89$	Not sig.
5 vs 4	$5.75 - 3.50 = 2.25 > 1.89$	Significant
2 vs 3	$4.75 - 4 = 0.75 < 1.89$	Not sig.
2 vs 4	$5.75 - 4 = 1.75 < 1.89$	//
3 vs 4	$5.75 - 4.75 = 1 < 1.89$	//

ii) Duncan's Multiple range test.

$$R_p = q_{\alpha}(p, v) \sqrt{\frac{MSE}{\delta}} \quad p = 2, 3, \dots, k-1, k.$$

Where $MSE = 1.57$.

δ (the number of observations in each treatment) = 4

v (df for MSE) = 15 and k (number of treatments) = 6

$$\rightarrow \text{Now: } \sqrt{\frac{MSE}{\delta}} = \sqrt{\frac{1.57}{4}} = 0.6265$$

→ Arranging the treatment means in ascending order of magnitude.

$\bar{X}_1$	$\bar{X}_6$	$\bar{X}_5$	$\bar{X}_2$	$\bar{X}_3$	$\bar{X}_4$
1.75	1.75	3.50	4.00	4.75	5.75

→ The values of $q_{\alpha, 0.05}(p, 15)$ for $p = 2, 3, 4, 5, 6$ taken from Duncan's table of significant ranges and the least significant ranges, R_p obtained by multiplying $q_{\alpha, 0.05}(p, 15)$ by $\sqrt{MSE/\delta}$, are shown below.

P	$q_{0.05}(P, 15)$	$q_{0.05}(P, 15)$	R_p
2	3.01		1.89 (R_2)
3	3.16		1.98 (R_3)
4	3.25		2.04 (R_4)
5	3.31		2.07 (R_5)
6	3.36		2.11 (R_6)

→ Comparing the differences b/w all pairs of means with the least significant ranges R_p , beginning with the largest ($\bar{x}_6$) against the smallest ($\bar{x}_1$), we have the following results.

4 vs 1	: $5.75 - 1.75 = 4.00$	$> 2.11 (R_6)$	Significant.
4 vs 6	: $5.75 - 1.75 = 4.00$	$> 2.07 (R_5)$	//
4 vs 5	: $5.75 - 3.50 = 2.25$	$> 2.04 (R_4)$	//
4 vs 2	: $5.75 - 4.00 = 1.75$	$< 1.98 (R_3)$	Not significant.
4 vs 3	: $5.75 - 4.75 = 1.00$	$< 1.89 (R_2)$	//
3 vs 1	: $4.75 - 1.75 = 3.00$	$> 2.07 (R_5)$	Significant.
3 vs 6	: $4.75 - 1.75 = 3.00$	$> 2.04 (R_4)$	//
3 vs 5	: $4.75 - 3.50 = 1.25$	$< 1.98 (R_3)$	Not significant.
3 vs 2	: $4.75 - 4.00 = 0.75$	$< 1.89 (R_2)$	//
2 vs 1	: $4.00 - 1.75 = 2.25$	$> 2.04 (R_4)$	Significant.
2 vs 6	: $4.00 - 1.75 = 2.25$	$> 1.98 (R_3)$	//
2 vs 5	: $4.00 - 3.50 = 0.50$	$< 1.89 (R_2)$	Not significant.
5 vs 1	: $3.50 - 1.75 = 1.75$	$< 1.98 (R_3)$	//
5 vs 6	: $3.50 - 1.75 = 1.75$	$< 1.89 (R_2)$	//
6 vs 1	: $1.75 - 1.75 = 0$	$< 1.89 (R_2)$	//

iii) Tukey's HSD = $q_{\alpha}(k, \text{error df}) \sqrt{MSE/n}$
 $\alpha = 0.05$, $k = 6$, $\text{error df} = 18$ that's error df.
 $n = 24$, $MSE = 1.57$.
 Tukey's HSD = $q_{0.05}(6, 18) \sqrt{1.57/24} = 4.59 \times \sqrt{1.57/24}$
 Tukey's HSD = 1.174
 $q_{\alpha}(k, \text{d.f. of error}) \sqrt{\frac{MSE}{n}}$

Arranging the treatment means in ascending order of their magnitude.

$\bar{X}_1$	$\bar{X}_6$	$\bar{X}_5$	$\bar{X}_2$	$\bar{X}_3$	$\bar{X}_4$	Proceede similar to LSP or
1.75	1.75	3.50	4.00	4.75	5.75	Duncans.

OR.

$\bar{X}_1$	$\bar{X}_2$	$\bar{X}_3$	$\bar{X}_4$	$\bar{X}_5$	$\bar{X}_6$
1.75	4.00	4.75	5.75	3.50	1.75

Now Comparing treatment means.

$$|\bar{X}_1 - \bar{X}_2| = |1.75 - 4.00| = 2.25 > 1.174 \text{ Significant.}$$

$$|\bar{X}_1 - \bar{X}_3| = 3 > 1.174 \text{ //}$$

$$|\bar{X}_1 - \bar{X}_4| = 4 > 1.174 \text{ //}$$

$$|\bar{X}_1 - \bar{X}_5| = 1.75 > 1.174 \text{ //}$$

$$|\bar{X}_1 - \bar{X}_6| = 0 < 1.174 \text{ Not significant.}$$

$$|\bar{X}_2 - \bar{X}_3| = 0.75 < 1.174 \text{ //}$$

$$|\bar{X}_2 - \bar{X}_4| = 1.75 > 1.174 \text{ Significant.}$$

$$|\bar{X}_2 - \bar{X}_5| = 0.50 < 1.174 \text{ Not significant.}$$

$$|\bar{X}_2 - \bar{X}_6| = 2.25 > 1.174 \text{ Significant.}$$

$$|\bar{X}_3 - \bar{X}_4| = 1 < 1.174 \text{ Not significant.}$$

$$|\bar{X}_3 - \bar{X}_5| = 1.25 > 1.174 \text{ Significant.}$$

$$|\bar{X}_3 - \bar{X}_6| = 3 > 1.174 \text{ //}$$

$$|\bar{X}_4 - \bar{X}_5| = 2.25 > 1.174 \text{ //}$$

$$|\bar{X}_4 - \bar{X}_6| = 4 > 1.174 \text{ //}$$

$$|\bar{X}_5 - \bar{X}_6| = 1.75 > 1.174 \text{ //}$$

v) Scheffe's Test:

To find critical value of Scheffe's Test:

$$F^* = (K-1) (\text{critical value of Anova})$$

$$F^* = (5)(2.90) \quad R^* =$$

$$F_s = \frac{(\bar{X}_i - \bar{X}_j)^2}{MSE \left(\frac{1}{n_i} + \frac{1}{n_j} \right)}$$

$$MSE \left(\frac{1}{n_i} + \frac{1}{n_j} \right)$$

$$\text{OR} \quad \frac{(\bar{X}_i - \bar{X}_j)^2}{\frac{2MSE}{n}}$$

$\bar{x}_1$ 1.75 $\bar{x}_2$ 4 $\bar{x}_3$ 4.75 $\bar{x}_4$ 5.75 $\bar{x}_5$ 3.5 $\bar{x}_6$ 1.75

$F = (k-1)(C.V. \text{ of Anova})$
 $F' = (5)(2.90)$
 $F' = 14.5$

Treatment	F_s	Conclusion:
1 vs 2	$\frac{(1.75-4)^2}{1.57(\frac{1}{4}+\frac{1}{4})} = 6.45 < 14.5$	not sig.
1 vs 3	11.64 < 14.5	not sig.
1 vs 4	20.38 > 14.5	significant
1 vs 5	3.90 < 14.5	not significant
1 vs 6	0 < 14.5	//
2 vs 3	0.71 < 14.5	//
2 vs 4	3.90 < 14.5	//
2 vs 5	0.32 < 14.5	//
2 vs 6	6.45 < 14.5	//
3 vs 4	1.27 < 14.5	//
3 vs 5	2 < 14.5	//
3 vs 6	11.46 < 14.5	//
4 vs 5	6.45 < 14.5	//
4 vs 6	20.38 > 14.5	significant
5 vs 6	3.90 < 14.5	not significant.